NETWORKWALKS B082

WEEK 2 PRACTICAL REPORT

FOOTPRINTING & RECONNAISSANCE

W2-PM1: Using Multiple Kali Linux Tools

Student	Morenike Jaiyesimi
Programme	NetworkWalks Cybersecurity Training
Batch	B082
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Target domain	networkwalks.com

Practical reconnaissance report documenting six tools, their observed outputs, analysis and evidence references.

1. Introduction

Footprinting and reconnaissance are important stages of cybersecurity assessment. They involve gathering information about a target to understand its publicly exposed infrastructure, technologies, DNS configuration and potential security exposure.

This practical exercise used six Kali Linux tools to conduct reconnaissance on the NetworkWalks domain, **networkwalks.com**. The work focused on understanding how each tool operates, interpreting its results and documenting findings for security analysis.

2. Objectives

- Retrieve publicly available domain registration information.
- Identify website technologies and server information.
- Resolve the domain name to its IP address.
- Inspect HTTP response headers.
- Identify the Web Application Firewall detected by the tool.
- Enumerate publicly available DNS records.
- Document findings using screenshots and saved command outputs.

3. Scope and Methodology

Target: networkwalks.com

The practical used the six tools specified in the NetworkWalks Week 2 PM1 assignment: WHOIS, WhatWeb, NSLOOKUP, CURL, WAFW00F and DNSRecon. Commands were executed in Kali Linux. Results were reviewed and documented. The exercise was limited to the assigned domain and publicly available information.

4. Practical Exercises and Findings

Task 1: WHOIS – Domain Registration Lookup

Objective: Retrieve publicly available domain registration information.

Command executed:

```
whois networkwalks.com
```

Findings:

- Domain: NETWORKWALKS.COM.
- Registrar: GoDaddy.com, LLC.
- Creation date: 6 November 2019; registry expiry date: 6 November 2027; updated date: 12 November 2025.
- Registrant information was protected through Domains By Proxy, LLC.
- Name servers shown across the returned registry/registrar information included NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM, NS29.DOMAINCONTROL.COM and NS30.DOMAINCONTROL.COM.
- DNSSEC: unsigned.

Analysis: The WHOIS lookup revealed domain registration details, registrar information, registration dates and name server information. The registrant's personal information was protected through a domain privacy service;

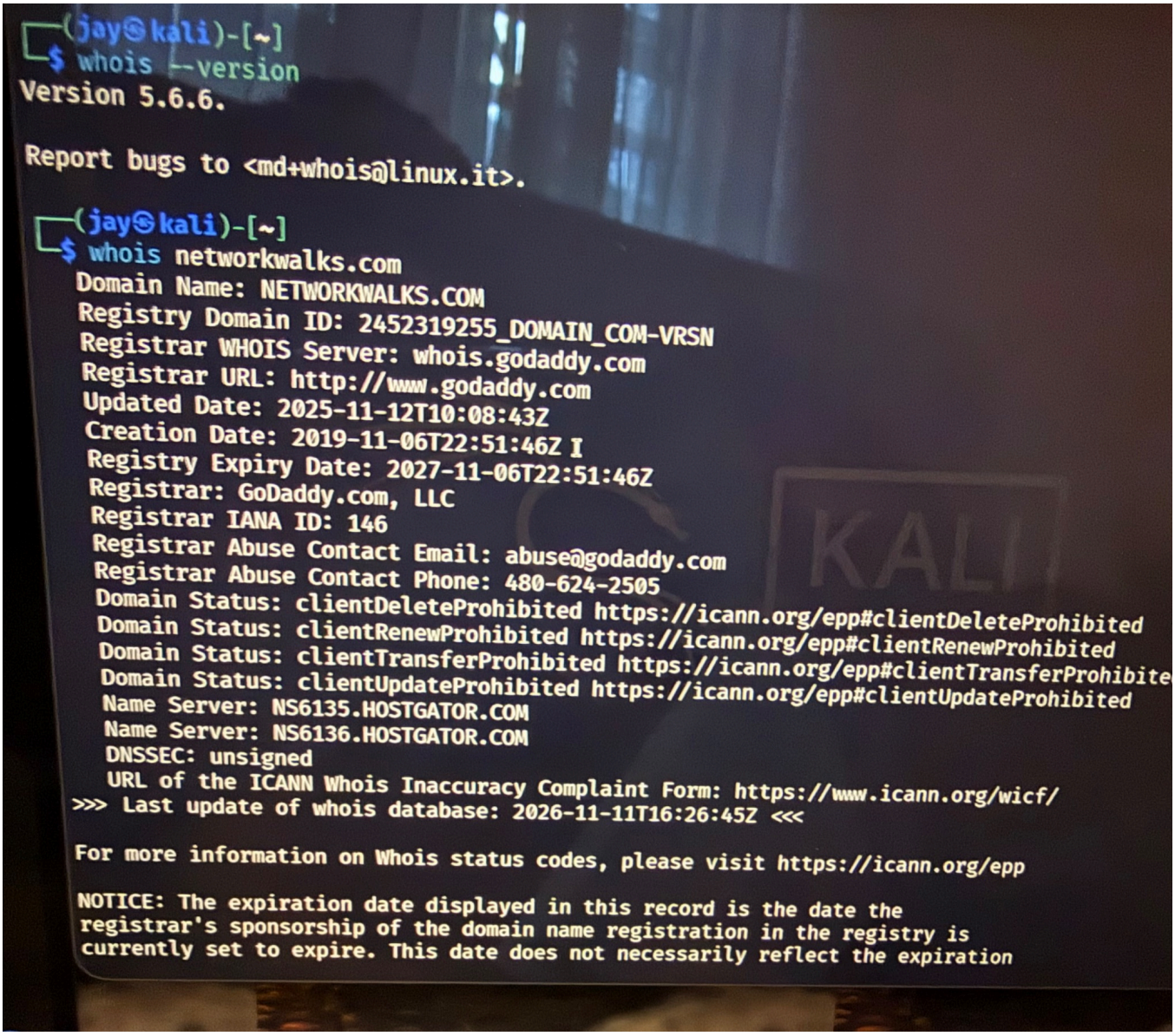
this is a privacy measure and does not indicate a vulnerability. The returned WHOIS information contained differences in the name server lists between registry and registrar views, so both were documented as returned.

Conclusion: The WHOIS lookup successfully provided useful information about the domain's registration and DNS administration.

Evidence files: PM1_Task1_WHOIS_Output.png, PM1_Task1_WHOIS_Output.txt

Task 1: WHOIS · Screenshot Evidence

Terminal output for the WHOIS lookup of networkwalks.com.



Task 2: WhatWeb – Website Technology Identification

Objective: Identify technologies and software associated with the target website.

Command executed:

```
whatweb networkwalks.com
```

Findings:

- Web server identified: Apache.
- Website platform reported: WordPress 7.1.4; WordPress Download Manager 3.3.76.
- Other detected technologies included Bootstrap 7.1.4, jQuery 3.7.1 and Google Tag Manager.
- Website title: Networkwalks Academy; public email detected: info@networkwalks.com.
- IP address reported: 192.232.216.135.
- HTTP returned 301 Moved Permanently and redirected to HTTPS; HTTPS returned 200 OK.

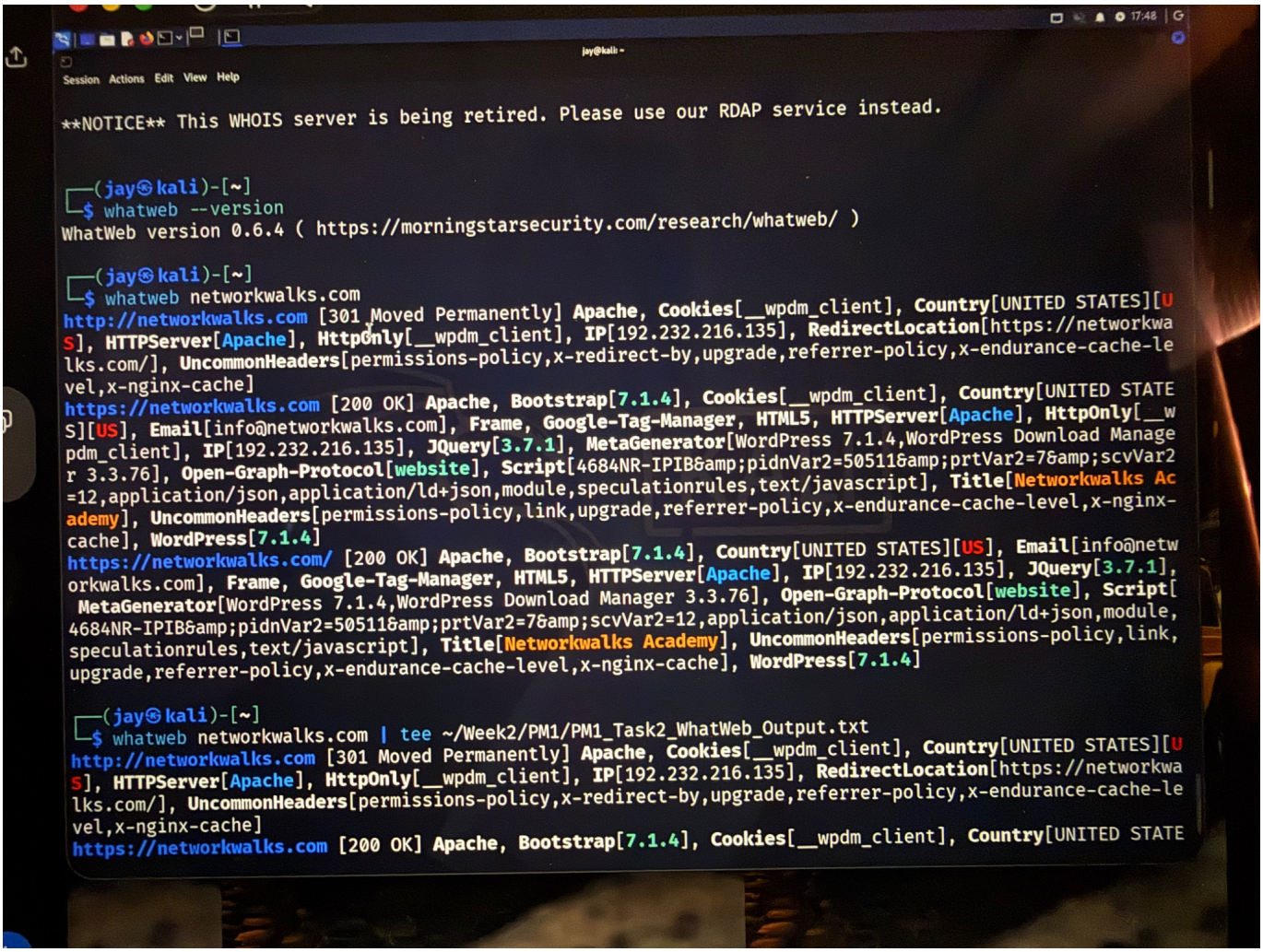
Analysis: WhatWeb identified several technologies associated with the website, including Apache, WordPress, Bootstrap and jQuery. The HTTP response redirected users to HTTPS, while the HTTPS request successfully returned the webpage. Technology version identification alone does not establish that a component is vulnerable.

Conclusion: WhatWeb successfully identified publicly detectable website technologies and provided useful information about the web infrastructure.

Evidence files: PM1_Task2_WhatWeb_Output.png, PM1_Task2_WhatWeb_Output.txt

Task 2: WhatWeb – Screenshot Evidence

Terminal output showing WhatWeb technology identification for networkwalks.com.



Task 3: NSLOOKUP – DNS Lookup

Objective: Identify the IP address associated with the domain.

Command executed:

```
nslookup networkwalks.com
```

Findings:

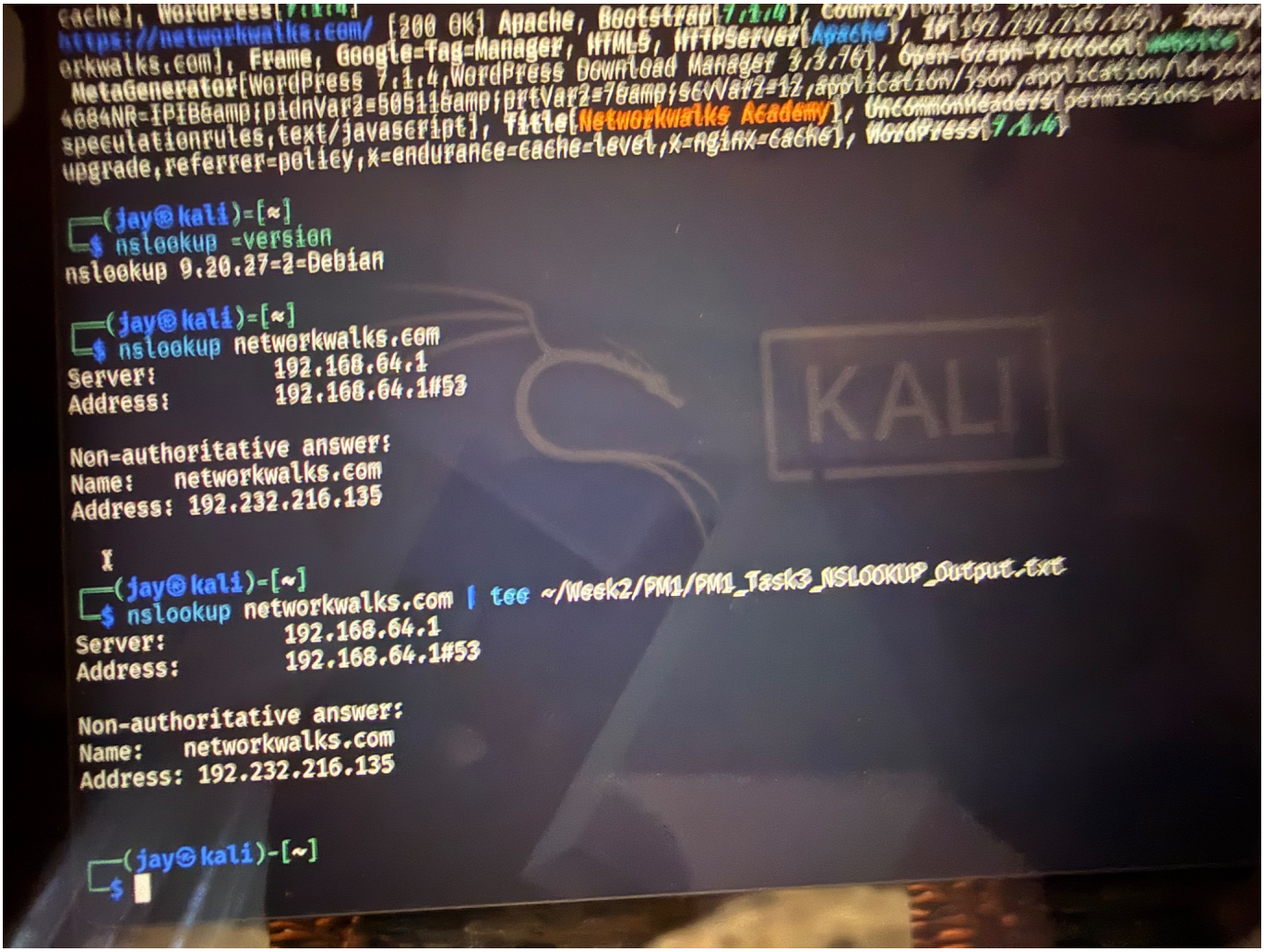
- DNS resolver: 192.168.64.1.
- DNS port: 53.
- Domain: networkwalks.com.
- Resolved IPv4 address: 192.232.216.135.
- Response type: non-authoritative answer.

Analysis: The command successfully resolved the domain name to its IPv4 address. The DNS resolver at 192.168.64.1 handled the request and returned 192.232.216.135. A non-authoritative answer indicates that the result was supplied by a DNS resolver rather than directly by the domain's authoritative name server.

Conclusion: The DNS lookup was successful and demonstrated how DNS translates domain names into IP addresses.

Task 3: NSLOOKUP – Screenshot Evidence

Terminal output showing DNS resolution for networkwalks.com.



Evidence files: PM1_Task3_NSLOOKUP_Output.png, PM1_Task3_NSLOOKUP_Output.txt

Task 4: CURL – HTTP Header Inspection

Objective: Inspect the HTTP response headers returned by the website.

Command executed:

```
curl -I https://networkwalks.com
```

Findings:

- HTTP response: HTTP/2 200.
- Server header: Apache.
- Content type: text/html; charset=UTF-8.
- WordPress REST API endpoints were advertised through /wp-json/ links.
- Cookie observed: __wpdm_client, with Secure and HttpOnly attributes.
- Referrer-Policy: no-referrer-when-downgrade; Permissions-Policy was also present.
- Cache-related headers included x-endurance-cache-level and x-nginx-cache.

Analysis: CURL successfully retrieved the HTTP response headers. The HTTP/2 200 response indicated that the server successfully processed the request. The response revealed WordPress REST API endpoints and a cookie containing Secure and HttpOnly attributes. Secure restricts cookie transmission to secure connections, while HttpOnly prevents direct JavaScript access to the cookie. These observations provide configuration indicators but do not by themselves prove that the website is secure or insecure.

Conclusion: The HTTP header inspection revealed useful information about the server response, WordPress endpoints, cookies and browser-related security policies.

Evidence files: PM1_Task4_CURL_Headers.png, PM1_Task4_CURL_Headers.txt

Task 5: WAFW00F – Web Application Firewall Detection

Objective: Identify whether a Web Application Firewall was detected on the target website.

Command executed:

```
wafw00f https://networkwalks.com
```

Findings:

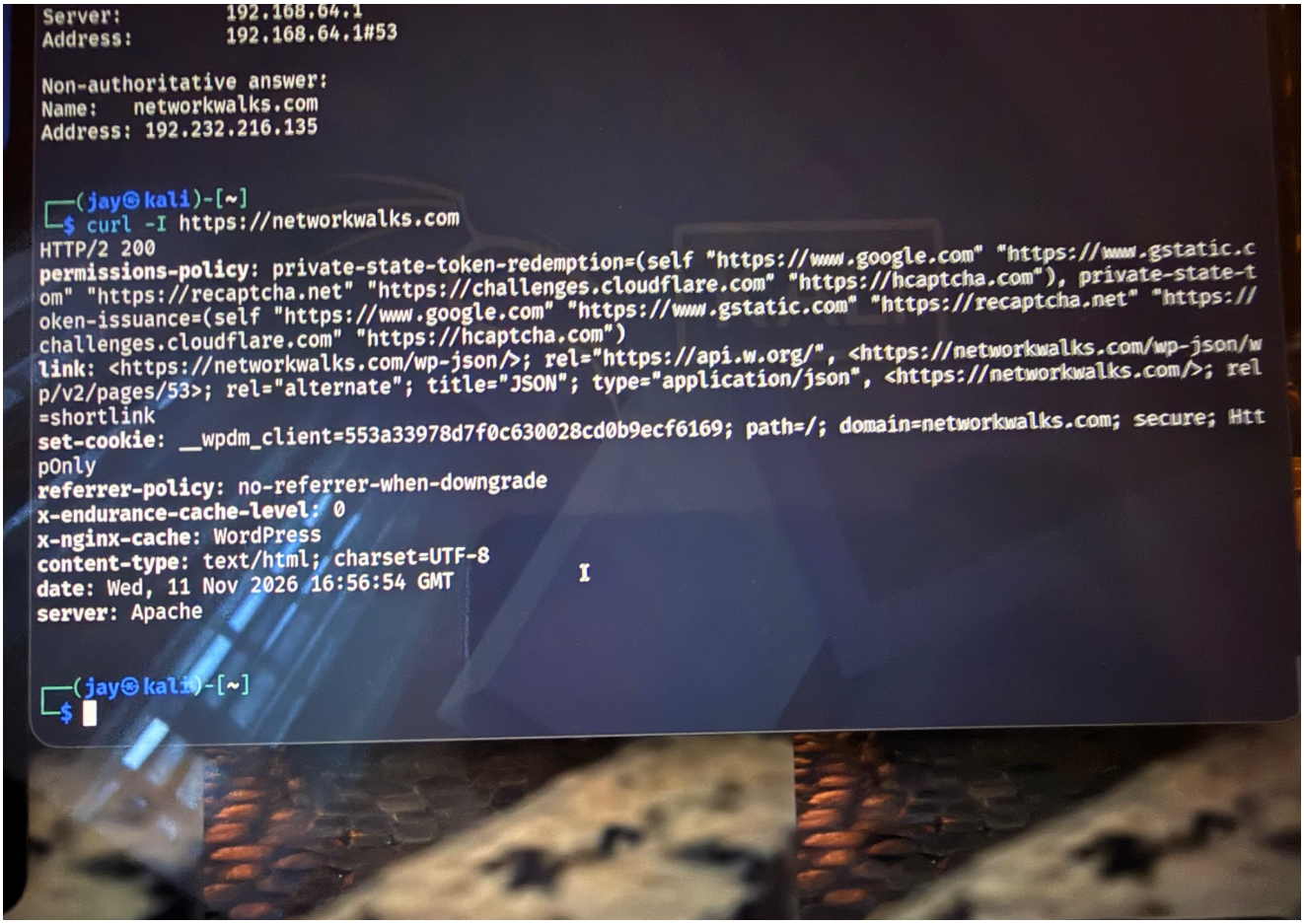
- WAFW00F version: 2.4.2.
- Target: https://networkwalks.com.
- Detected WAF: ModSecurity (SpiderLabs).
- Number of requests: 2.
- Detection completed successfully.

Analysis: WAFW00F identified ModSecurity (SpiderLabs) as the Web Application Firewall associated with the target website. ModSecurity is an open-source WAF designed to inspect incoming web requests and help detect or block potentially malicious HTTP traffic. The detection does not establish the firewall's exact configuration, effectiveness or whether all protection rules are enabled.

Conclusion: The WAF detection exercise identified ModSecurity (SpiderLabs) as the firewall technology detected on the target website.

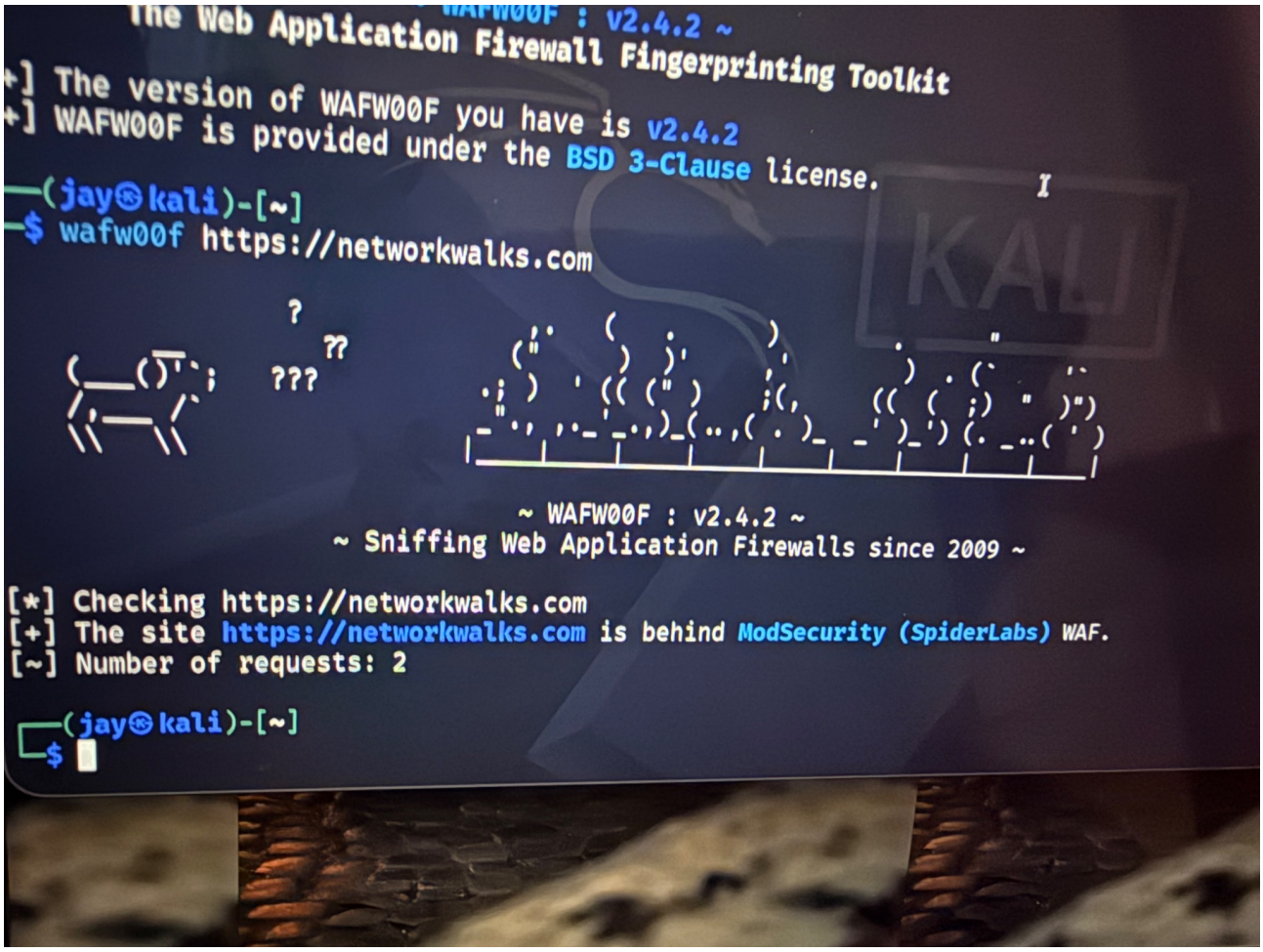
Task 4: CURL – Screenshot Evidence

Terminal output showing HTTP response headers for https://networkwalks.com.



Task 5: WAFW00F – Screenshot Evidence

Terminal output showing WAFW00F detection for https://networkwalks.com.



Evidence files: PM1_Task5_WAFW00F_Output.png, PM1_Task5_WAFW00F_Output.txt

Task 6: DNSRecon – DNS Enumeration

Objective: Identify publicly available DNS records and gather information about the domain's DNS infrastructure.

Command executed:

```
dnsrecon -d networkwalks.com
```

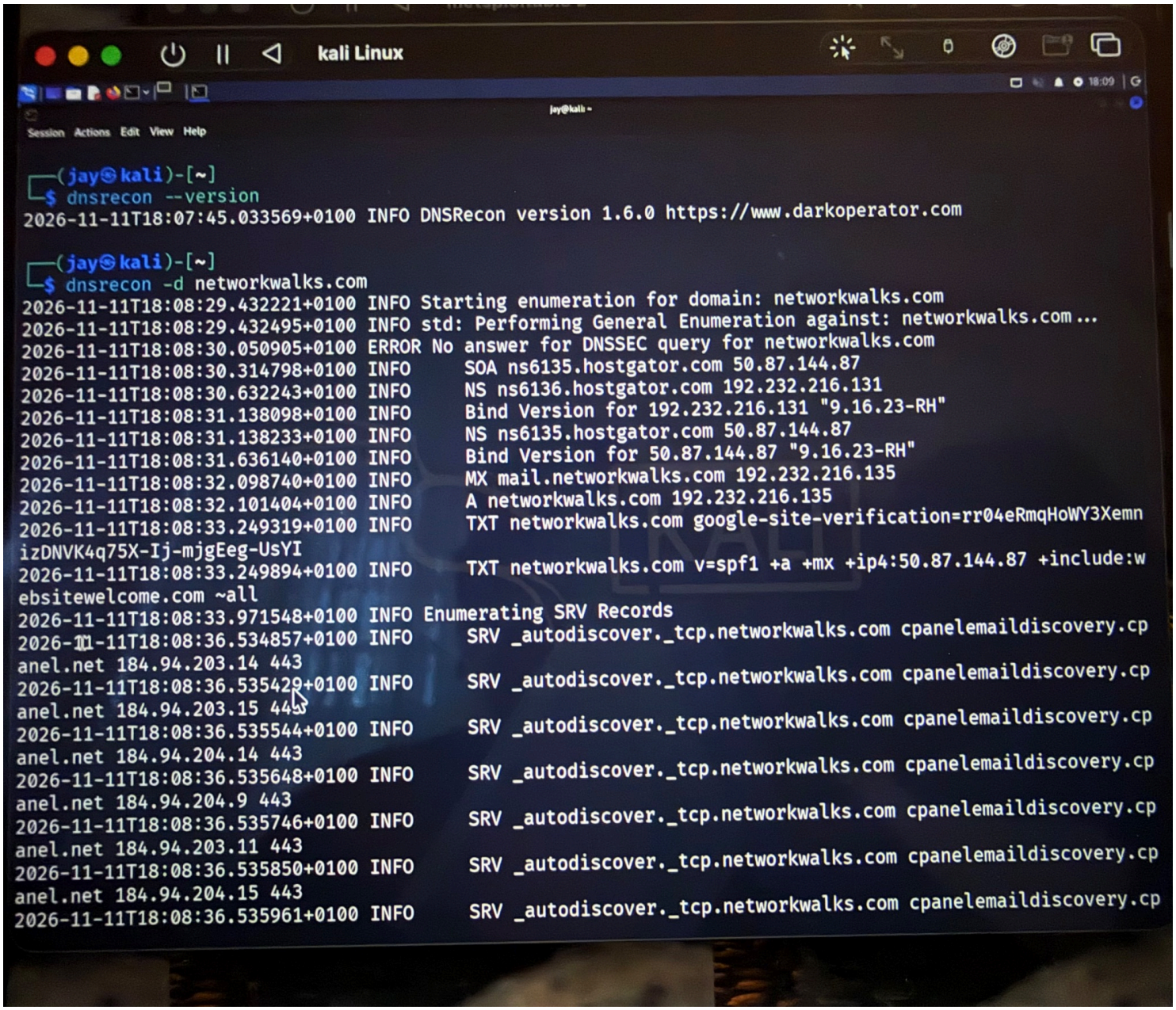
Findings:

- DNSRecon version: 1.6.0; enumeration completed.
- SOA: ns6135.hostgator.com (50.87.144.87).
- Name servers: ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131).
- BIND version information disclosed by both name servers: 9.16.23-RH.
- MX hostname: mail.networkwalks.com, resolving in the output to 192.232.216.135.
- A record: networkwalks.com → 192.232.216.135.
- TXT records included a Google site-verification value and an SPF policy.
- Eight SRV autodiscovery records were listed; DNSRecon reported eight records found.
- DNSSEC query returned no answer.

Analysis: DNSRecon identified DNS records associated with the target domain. SOA and NS records provided information about DNS administration; the MX record identified the mail hostname; and the A record resolved the domain to its IPv4 address. The TXT records revealed a Google verification entry and an SPF policy ending in ~all, which represents a soft-fail treatment for sources not otherwise authorised by the policy. SRV records revealed email autodiscovery information. The DNSSEC query result is consistent with the unsigned DNSSEC status observed in WHOIS. Disclosed version information and DNS records are reconnaissance findings, not proof of exploitable vulnerabilities.

Conclusion: DNSRecon successfully completed enumeration and revealed information about the target's DNS infrastructure, mail configuration and service records.

Evidence files: PM1_Task6_DNSRecon_Output.png, PM1_Task6_DNSRecon_Output.txt


kali Linux
Session Actions Edit View Help
(jay㉿kali)-[~]
$ dnsrecon --version
2026-11-11T18:07:45.033569+0100 INFO DNSRecon version 1.6.0 https://www.darkoperator.com
(jay㉿kali)-[~]
$ dnsrecon -d networkwalks.com
2026-11-11T18:08:29.432221+0100 INFO Starting enumeration for domain: networkwalks.com
2026-11-11T18:08:29.432495+0100 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-11-11T18:08:30.050905+0100 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-11T18:08:30.314798+0100 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-11-11T18:08:30.632243+0100 INFO NS ns6136.hostgator.com 192.232.216.131
2026-11-11T18:08:31.138098+0100 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-11T18:08:31.138233+0100 INFO NS ns6135.hostgator.com 50.87.144.87
2026-11-11T18:08:31.636140+0100 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-11T18:08:32.098740+0100 INFO MX mail.networkwalks.com 192.232.216.135
2026-11-11T18:08:32.101404+0100 INFO A networkwalks.com 192.232.216.135
2026-11-11T18:08:33.249319+0100 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3Xemn
izDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-11T18:08:33.249894+0100 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:w
ebsitewelcome.com ~all
2026-11-11T18:08:33.971548+0100 INFO Enumerating SRV Records
2026-11-11T18:08:36.534857+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.14 443
2026-11-11T18:08:36.535429+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.15 443
2026-11-11T18:08:36.535544+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.14 443
2026-11-11T18:08:36.535648+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.9 443
2026-11-11T18:08:36.535746+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.203.11 443
2026-11-11T18:08:36.535850+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp
anel.net 184.94.204.15 443
2026-11-11T18:08:36.535961+0100 INFO SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cp

5. Summary of Findings

Tool	Main finding
WHOIS	Registrar, registration dates, privacy service and name server information.
WhatWeb	Apache, WordPress and associated website technologies.
NSLOOKUP	Domain resolved to 192.232.216.135.
CURL	HTTP response headers, WordPress API links and cookie attributes.
WAFW00F	ModSecurity (SpiderLabs) detected.
DNSRecon	DNS, mail, TXT and SRV records enumerated.

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6. Lessons Learned

- Used WHOIS to retrieve publicly available domain registration information.
- Used WhatWeb to identify website technologies.
- Understood DNS resolution and the relationship between domain names and IP addresses.
- Inspected HTTP response headers using CURL.
- Learned the role of Web Application Firewalls and used WAFW00F to identify a detected WAF.
- Used DNSRecon to enumerate DNS records and interpret mail and service information.
- Practised saving terminal outputs and screenshots as evidence of cybersecurity work.
- Learned to distinguish reconnaissance observations from confirmed security vulnerabilities.

7. Overall Conclusion

The Week 2 PM1 Footprinting and Reconnaissance practical was completed using six Kali Linux tools. The exercise provided practical exposure to domain information gathering, website technology identification, DNS resolution, HTTP header analysis, firewall detection and DNS enumeration.

The findings demonstrated how publicly available information can help cybersecurity professionals understand a website's infrastructure and identify areas that may require further security assessment. The six practical exercises were documented with terminal outputs and screenshot evidence.

Evidence register

The following evidence files were created in the Kali Linux PM1 working folder. The screenshots are referenced by filename in this report; they are not embedded in this PDF because the image files remained inside the user's Kali virtual machine during report generation.

Task	Screenshot	Text output
Task 1	PM1_Task1_WHOIS_Output.png	PM1_Task1_WHOIS_Output.txt
Task 2	PM1_Task2_WhatWeb_Output.png	PM1_Task2_WhatWeb_Output.txt
Task 3	PM1_Task3_NSLOOKUP_Output.png	PM1_Task3_NSLOOKUP_Output.txt
Task 4	PM1_Task4_CURL_Headers.png	PM1_Task4_CURL_Headers.txt
Task 5	PM1_Task5_WAFW00F_Output.png	PM1_Task5_WAFW00F_Output.txt
Task 6	PM1_Task6_DNSRecon_Output.png	PM1_Task6_DNSRecon_Output.txt

Report prepared for the NetworkWalks B082 Week 2 PM1 practical submission.